Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates, with at least one term correct	AO1.1a	M1	$\frac{dv}{dt} = 12t - 36^2$
	Selects and applies $F = ma$ to 'their' derivative Condone use of 400 for mass	AO1.1a	M1	$F = ma = 0.4\,(12t - 36t^2)$
	Obtains correct expression for force FT from 'their' $F = ma$ equation, provided the first M1 has been awarded (may be in factorised form)	AO1.1b	A1F	$= 4.8t - 14.4t^2$
(b)	Integrates v to find r, with at least one term correct	AO3.1b	M1	$r = \int (6t^2 - 12t^3)\,dt$
	Obtains correct integral (condone absence of c)	AO1.1b	A1	$r = 2t^3 - 3t^4 + c$
	Deduces the value of c using initial conditions FT use of 'their' integral provided M1 awarded	AO2.2a	A1F	When $t = 0$, $r = 0$ so $0 = 2 \times 0^2 - 3 \times 0^4 + c$ so $c = 0$
	Forms and solves an equation for t (condone numerical slip)	AO1.1a	M1	$0 = 2t^3 - 3t^4$ $= t^3(2 - 3t)$ $t = 0$ or $t = \frac{2}{3}$
	Interprets solution, realising that the non-zero time is required (Must include units) FT use of 'their' equation for t provided both M1 marks have been awarded	AO3.2a	A1F	Next at O at $\frac{2}{3}$ seconds
				Total 8 marks